

Q2)

Known

Closed system  
Polytropic Expansion  
 $\bar{R} = 8.31434 \text{ kJ/kmol}\cdot\text{K}$   
 $M_{\text{CO}_2} = 44.01 \text{ kg/kmol}$   
 $T_i = 500^\circ\text{K}$

Assumptions

$\text{CO}_2 \rightarrow \text{I.G.}$

Polytropic Expansion  
 $\Rightarrow \Delta\psi$

i.e. Net change/  
change of state

Neglect  $\Delta PE + \Delta KE$

Diagram

initial State

$\text{CO}_2$   
 $T_i = 600^\circ\text{K}$   
 $P_i = 8 \text{ MPa}$

final State

$\text{CO}_2$   
 $T_f = 350^\circ\text{K}$   
 $P_f = ?$

$n = 1.1$

Determine

$P_f$

$w_{b/m} = w_b$

$\sigma_m$

Symbolic

C.O.M.:  $m_f - m_i = 0 \Rightarrow m_f = m_i$

1<sup>st</sup> Law C.O.E.:  $E_f - E_i = Q - W$

2<sup>nd</sup> Law:  $S_f - S_i = \Sigma \frac{Q}{T_b} + \sigma$

$\frac{T_f}{T_i} = \left(\frac{P_f}{P_i}\right)^{\frac{n-1}{n}} \Rightarrow P_f = P_i \left(\frac{T_f}{T_i}\right)^{\frac{n}{n-1}}$

$w_b = \frac{R(T_f - T_i)}{1-n}$

$E_f - E_i = Q - W$

$\Rightarrow u_f - u_i = Q - w_b \quad * u = \bar{u}$

$\Rightarrow u_f - u_i = q - w_b \Rightarrow q = u_f - u_i + w_b$

$\Rightarrow q = \frac{\bar{u}_f - \bar{u}_i}{M_{\text{CO}_2}} + w_b$

$\kappa = \frac{\bar{R}}{M_{\text{CO}_2}}$

from table A-23

@  $T_i$   $\bar{u}_i = 17291 \text{ kJ/kmol}$   
 $\bar{s}_i = 243.149 \text{ kJ/kmol}\cdot\text{K}$   
@  $T_f$   $\bar{u}_f = 8439 \text{ kJ/kmol}$   
 $\bar{s}_f = 219.831 \text{ kJ/kmol}\cdot\text{K}$

For I.G. Change of State  $\Rightarrow s_f - s_i = s_f^* - s_i^* - R \ln\left(\frac{P_f}{P_i}\right)$

Polytropic I.G.  $|n \neq 1$

$\Rightarrow w_b = \frac{R(T_f - T_i)}{1-n} + \frac{T_f}{T_i} \left(\frac{P_f}{P_i}\right)^{\frac{n-1}{n}}$

$S_f - S_i = \Sigma \frac{Q}{T_b} + \sigma$

$s_f - s_i = \Sigma \frac{q}{T_b} + \sigma_m \quad * s^* = \bar{s}$

$\Rightarrow s_f - s_i - \frac{q}{T_b} = \sigma_m \Rightarrow \sigma_m = s_f - s_i - R \ln\left(\frac{P_f}{P_i}\right) - \frac{q}{T_b}$

$\Rightarrow \sigma_m = \frac{\bar{s}_f - \bar{s}_i}{M_{\text{CO}_2}} - R \ln\left(\frac{P_f}{P_i}\right) - \frac{q}{T_b}$

### Numerical Solutions:

$$P_s = P_i \left( \frac{T_s}{T_i} \right)^{\frac{n}{n-1}} = 8 \text{ MPa} \left( \frac{350 \text{ K}}{600 \text{ K}} \right)^{\frac{1.1}{1.1-1}} = 0.02129 \text{ MPa}$$

$$P_s = 21.29 \text{ kPa}$$

$$R = \frac{\bar{R}}{M_{\text{CO}_2}} = \frac{8.31434 \text{ kJ/kmol} \cdot \text{K}}{44.01 \text{ kg/kmol}} = 0.18892 \text{ kJ/kg} \cdot \text{K}$$

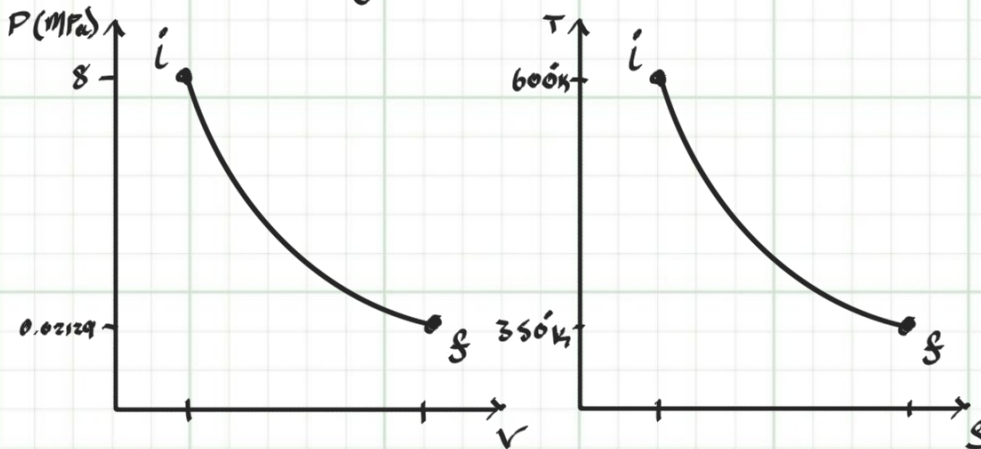
$$w_b = \frac{0.18892 \text{ kJ/kg} \cdot (350 \text{ K} - 600 \text{ K})}{1 - (1.1)} = w_b = 472.3 \text{ kJ/kg}$$

$$q = \frac{\bar{u}_i - \bar{u}_s}{M_{\text{CO}_2}} + w_b = \frac{(8439 - 17291) \text{ kJ/kmol}}{44.01 \text{ kg/kmol}} + 472.3 \text{ kJ/kg} = q = 271.16 \text{ kJ/kg}$$

$$\sigma_m = \frac{\bar{s}_i - \bar{s}_s}{M_{\text{CO}_2}} - R \ln \left( \frac{P_s}{P_i} \right) - \frac{q}{T_b} = \frac{(219.831 - 243.140) \text{ kJ/kmol}}{44.01 \text{ kg/kmol}} - (0.18892 \text{ kJ/kg} \cdot \text{K}) \ln \left( \frac{0.02129 \text{ MPa}}{8 \text{ MPa}} \right) - \frac{271.16 \text{ kJ/kg}}{500 \text{ K}}$$

$$\sigma_m = 0.0468 \text{ kJ/kg} \cdot \text{K}$$

### P-V + T-s Diagrams





Q3)

Determine

$\sigma$

$$T_g = T_{H_2O_f} = T_{Al_f}$$

Known

From Table A-19

$$Al @ 360^\circ K \begin{cases} c_p = 0.903 \text{ kJ/kg}^\circ K \\ \rho = 2700 \text{ kg/m}^3 \\ k = 237 \text{ W/m}^\circ K \end{cases}$$

$$H_2O @ 325^\circ K \begin{cases} c_p = 4.182 \text{ kJ/kg}^\circ K \\ \rho = 987.1 \text{ kg/m}^3 \\ k = 0.645 \text{ W/m}^\circ K \end{cases}$$

Assumptions

Closed system

- Net change

Rigid Tank

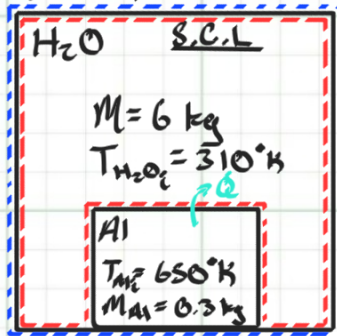
$\Rightarrow$  Isometric i.e.  $W=0$

$\Rightarrow$  Adiabatic

$$\Delta E_{sys} = 0$$

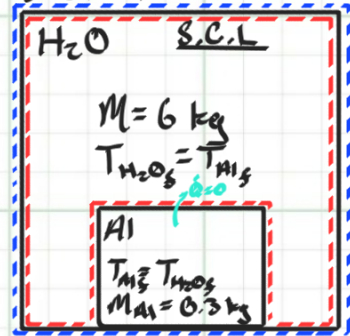
Diagram

initial state



well insulated

final state



Symbolic

general form:  $\Delta E = Q - W$

$$\Delta E_{sys} = \Delta E_{Al} + \Delta E_{H_2O} = 0$$

$$\Delta E_{sys} = (E_{Al_f} - E_{Al_i}) + (E_{H_2O_f} - E_{H_2O_i}) = 0$$

$$\Delta S_{sys} = \Delta S_{H_2O} + \Delta S_{Al} = \sum \frac{Q}{T_b} + \sigma$$

$$\Delta S_{sys} = (S_{H_2O_f} - S_{H_2O_i}) + (S_{Al_f} - S_{Al_i}) = \sigma$$

\* Note:  $H_2O$  stays in subcool liquid region

$\left\{ \begin{array}{l} \bullet \bullet \text{ Acting as an Incompressible Refrigerant} \\ \bullet \bullet \text{ Al block is a solid} \end{array} \right.$

$\left\{ \begin{array}{l} \bullet \bullet \text{ For All incompressible substances} \\ S_2 - S_1 = C \ln\left(\frac{T_2}{T_1}\right) = \Delta S \\ u_2 - u_1 = C(T_2 - T_1) = \Delta u \end{array} \right.$

$$\therefore \Delta E = \Delta U = \Delta u m = m c (T_2 - T_1)$$

$$\Delta S = \Delta s u = m c \ln\left(\frac{T_2}{T_1}\right) \quad \left\{ \begin{array}{l} \text{The above 2 general equations are the same for both} \\ H_2O \text{ \& \;} Al \text{ but are property specific to the substance} \end{array} \right.$$

$$0 = \Delta E_{sys} = M_{Al} c_{Al} (T_g - T_{Al_i}) + M_{H_2O} c_{H_2O} (T_g - T_{H_2O_i})$$

$$\Rightarrow M_{Al} c_{Al} (T_{Al_i} - T_g) = M_{H_2O} c_{H_2O} (T_g - T_{H_2O_i}) \Rightarrow \frac{M_{Al} c_{Al} T_{Al_i} + M_{H_2O} c_{H_2O} T_{H_2O_i}}{M_{H_2O} c_{H_2O} + M_{Al} c_{Al}} = T_g$$

$$\Delta S_{sys} = M_{Al} c_{Al} \ln\left(\frac{T_g}{T_{Al_i}}\right) + M_{H_2O} c_{H_2O} \ln\left(\frac{T_g}{T_{H_2O_i}}\right) = \sigma$$

Numerical Solution:

$$T_g = \frac{m_{H_2O} c_{H_2O} T_{H_2O,i} + m_{H_2O} c_{H_2O} T_{H_2O,i}}{m_{H_2O} c_{H_2O} + m_{H_2O} c_{H_2O}} = \frac{(0.3 \text{ kg})(0.903 \frac{\text{kJ}}{\text{kg}\cdot\text{K}})(150^\circ\text{K}) + (6 \text{ kg})(4.182 \frac{\text{kJ}}{\text{kg}\cdot\text{K}})(310^\circ\text{K})}{(0.3 \text{ kg})(0.903 \frac{\text{kJ}}{\text{kg}\cdot\text{K}}) + (6 \text{ kg})(4.182 \frac{\text{kJ}}{\text{kg}\cdot\text{K}})}$$

$$T_g = 313.63^\circ\text{K}$$

$$\Delta S_{\text{sys}} = m_{H_2O} c_{H_2O} \ln\left(\frac{T_g}{T_{H_2O,i}}\right) + m_{H_2O} c_{H_2O} \ln\left(\frac{T_g}{T_{H_2O,i}}\right) = 0$$

$$= (0.3 \text{ kg})(0.903 \frac{\text{kJ}}{\text{kg}\cdot\text{K}}) \ln\left(\frac{313.63^\circ\text{K}}{650^\circ\text{K}}\right) + (6 \text{ kg})(4.182 \frac{\text{kJ}}{\text{kg}\cdot\text{K}}) \ln\left(\frac{313.63^\circ\text{K}}{310^\circ\text{K}}\right)$$

$$\Delta S_{\text{sys}} = 0.09469 \frac{\text{kJ}}{\text{K}} = 0$$



(Q4)

Determine

$$\dot{M}_{H_2O}$$

Assumptions

Isobaric

Steady State, Open System

Neglect  $\Delta PE$   $\Delta KE$

Adiabatic

I.G. for Air

Known

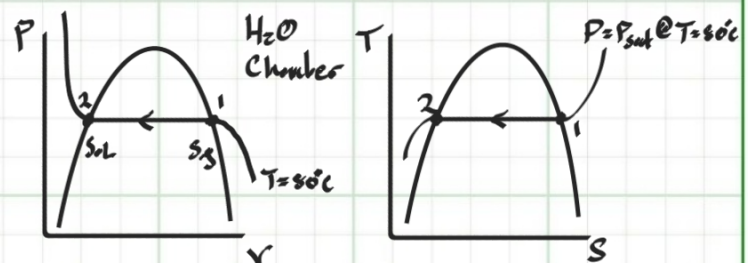
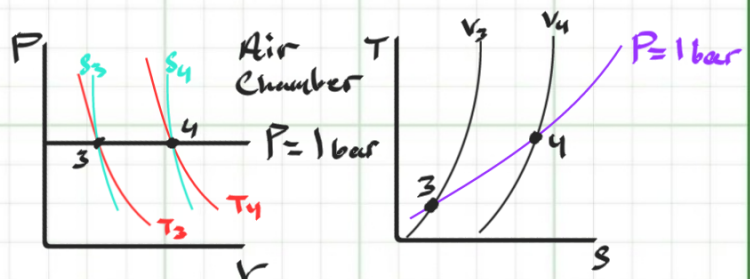
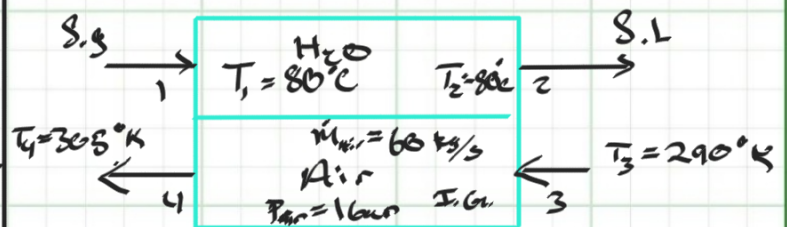
$$T_1 = T_2 = 80^\circ C$$

$$\dot{M}_{air} = \dot{M}_3 = \dot{M}_4 = 60 \text{ kg/s}$$

$$T_3 = 290^\circ K \quad T_4 = 305^\circ K$$

@ State 1 S.g @ State 2 S.L

Diagram



Symbolic Solution

for Air chamber

C.O.M:  $\dot{M}_3 = \dot{M}_4 = 60 \text{ kg/s}$

C.O.E:  $0 = \dot{Q} - \dot{W} + \dot{M}_3(h_3 - h_4)$

$$\Rightarrow \dot{W}_4 = \dot{M}_4(h_3 - h_4)$$

∴ for I.G.  $\Delta h = c_p \Delta T$  |  $c_p = c_{p,air}$

$$\Rightarrow \dot{W}_4 = \dot{M}_4 c_{p,air} (T_3 - T_4)$$

\* Note: for  $\dot{W}_6$  between Air &  $H_2O$  chambers must be equivalent

i.e.  $\dot{W}_{6,air} = \dot{W}_2 = \dot{W}_4$

for  $H_2O$  chamber

C.O.M:  $\dot{M}_1 = \dot{M}_2 \Rightarrow \dot{M}_1 = \dot{M}_2 = \dot{M}_2$

C.O.E:  $0 = \dot{Q} - \dot{W}_2 + \dot{M}_1(h_1 - h_2)$

$$\Rightarrow \dot{W}_2 = \dot{M}_2(h_1 - h_2)$$

∴ State points are @ S.L + S.g

$$\Rightarrow \{h_1 = h_g, h_2 = h_g\} | T = 80^\circ C$$

$$\Rightarrow \dot{W}_2 = \dot{M}_2(h_{g,80^\circ C} - h_{s,80^\circ C})$$

$$\Rightarrow \dot{M}_2 = \frac{\dot{M}_4 c_{p,air} (T_3 - T_4)}{(h_{g,80^\circ C} - h_{s,80^\circ C})}$$

2<sup>nd</sup> Law:  $0 = \cancel{\dot{Q}}_{12} + \sum \dot{m}_i s_i - \sum \dot{m}_e s_e + \dot{\sigma}$

$$\Rightarrow \dot{\sigma} = \dot{m}_2 (s_2 - s_1) + \dot{m}_4 (s_4 - s_3)$$

$$\because \text{For I.G., } \Delta s = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right) \Big|_{C_p = C_{p, \text{air}}}$$

& for the s.g. & s.l. state points for H<sub>2</sub>O

$$\{s_1 = s_3, s_2 = s_3\} \mid T = 80^\circ\text{C}$$

$$\therefore \dot{\sigma} = \dot{m}_2 (s_{g, 80^\circ\text{C}} - s_{g, 80^\circ\text{C}}) + \dot{m}_4 C_{p, \text{air}} \ln\left(\frac{T_4}{T_3}\right) - R \ln\left(\frac{P_4}{P_3}\right) \quad \text{0: Isobaric}$$

$$\boxed{\dot{\sigma} = \dot{m}_2 (s_{g, 80^\circ\text{C}} - s_{g, 80^\circ\text{C}}) + \dot{m}_4 C_{p, \text{air}} \ln\left(\frac{T_4}{T_3}\right)}$$

Numerical Solution:

$$\because C_{p, \text{air}} = C_p @ T_{\text{avg, air}} \mid T_{\text{avg, air}} \approx 300^\circ\text{K}$$

$$\therefore C_{p, \text{air}} \approx 1.005$$

@ T = 80°C for Air

$$h_{g, 80^\circ\text{C}} = 334.91 \text{ kJ/kg}$$

$$h_{g, 80^\circ\text{C}} = 2643.7 \text{ kJ/kg}$$

$$s_{g, 80^\circ\text{C}} = 1.0783 \text{ kJ/kg}\cdot\text{K}$$

$$s_{g, 80^\circ\text{C}} = 7.6122 \text{ kJ/kg}\cdot\text{K}$$

$$\dot{m}_2 = \frac{\dot{m}_4 C_{p, \text{air}} (T_3 - T_4)}{(h_{g, 80^\circ\text{C}} - h_{g, 80^\circ\text{C}})}$$

$$\Rightarrow \frac{60 \text{ kg/s} (1.005 (-15)) \text{ kJ/kg}}{(334.91 - 2643.7) \text{ kJ/kg}}$$

$$\Rightarrow \boxed{\dot{m}_2 = 0.3918 \text{ kg/s}}$$

$$\dot{\sigma} = \dot{m}_2 (s_{g, 80^\circ\text{C}} - s_{g, 80^\circ\text{C}}) + \dot{m}_4 C_{p, \text{air}} \ln\left(\frac{T_4}{T_3}\right)$$

$$\dot{\sigma} = (0.3918 \text{ kg/s}) (1.0783 - 7.6122) \text{ kJ/kg}\cdot\text{K} + 60 \text{ kg/s} (1.005 \ln\left(\frac{305}{290}\right)) \text{ kJ/kg}\cdot\text{K}$$

$$\boxed{\dot{\sigma} = 0.4801 \text{ kJ/s}\cdot\text{K} = \text{kW/K}}$$



Q5)

Determine

$$\dot{W}/m$$

$$\dot{S}/m$$

Assumptions

Adiabatic  $\because$  well insulated

Open System, S.S.

Neglect  $\Delta PE$   $\Delta KE$

Known

I.C. @ Both states

$\because$  Properties @ Both states  $\Rightarrow$  S.H.V.

From Table A-4

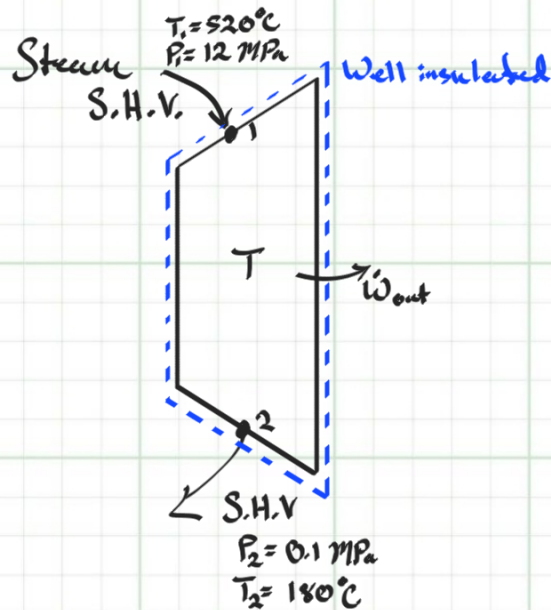
$$h_1 = 3401.8 \text{ kJ/kg}$$

$$h_2 = 2835.75 \text{ kJ/kg}$$

$$s_1 = 6.555 \text{ kJ/kg}\cdot\text{K}$$

$$s_2 = 7.747 \text{ kJ/kg}\cdot\text{K}$$

Diagram



Symbolic

C.O.M:  $\dot{m}_1 = \dot{m}_2 = m$

C.O.E:  $0 = \cancel{Q} - \dot{W} + \dot{m}_1 h_1 - \dot{m}_2 h_2$

$$\Rightarrow \dot{W} = m(h_1 - h_2)$$

$$\Rightarrow \frac{\dot{W}}{m} = (\dot{W} = h_1 - h_2)$$

2<sup>nd</sup> Law:  $0 = \cancel{\frac{Q}{T_b}} + m(s_1 - s_2) + \dot{S}$

$$\Rightarrow \dot{S} = m(s_2 - s_1)$$

$$\Rightarrow \left( \frac{\dot{S}}{m} = s_2 - s_1 \right)$$

Numerical Solutions:

$$\dot{W} = h_1 - h_2$$

$$\dot{W} = 3401.8 \text{ kJ/kg} - 2835.75 \text{ kJ/kg}$$

$$(\dot{W} = 566.1 \text{ kJ/kg})$$

$$\dot{S}/m = s_2 - s_1$$

$$\dot{S}/m = 7.747 \text{ kJ/kg}\cdot\text{K} - 6.555 \text{ kJ/kg}\cdot\text{K}$$

$$(\dot{S}/m = 1.192 \text{ kJ/kg}\cdot\text{K})$$